

COMBINED SCIENCE

FORM ONE



HAZARDS THAT COULD OCCUR DURING PRACTICAL WORK



SOME OF THE HAZARDS IN A LABORATORY

8.1.1 LABORATORY SAFETY AND APPARATUS

EXPLAIN LABORATORY RULES

- The equipment and chemical substances used in the laboratory are safe if used properly;
- To ensure that the laboratory is safe to work in, it is **very important** to obey the following rules:

LABORATORY RULES

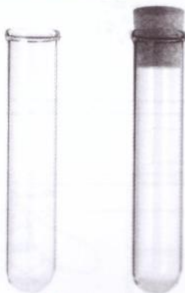
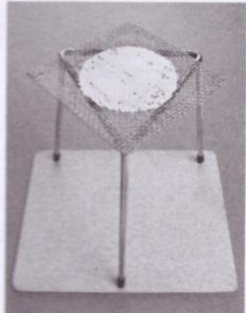
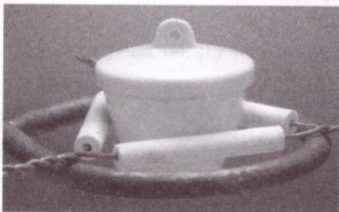
- Follow **all** instructions carefully;
- Consider the safety of other learners during practical work;
- Do not work in the laboratory without your teacher;
- Only do practical work that has been assigned to you by your teacher;
- Do not use any chemical until the teacher has explained its use and the precautions to be taken;
- Do not touch materials or equipment on the demonstration table without instructions from your teacher;
- Know where you can find the fire extinguisher and first-aid materials, and know how to use them;
- Always be careful when pouring chemicals from their containers;
- Report any spill to your teacher;
- If chemicals come into contact with your eyes or skin. You should wash the area immediately with running water;
- Never look into the mouth of a test tube whilst heating liquids in it, and never point the mouth of the tube in the direction of other learners;
- Gas burners must be tightly closed after use;
- Ensure that the laboratory is well **ventilated** when working in it;
- Wear lab laboratory coat and safety glasses whenever necessary;
- When using **volatile** and **flammable** liquids, such as alcohol, no flame should be used;
- Large storage bottles with dangerous chemicals should be handled only by the teacher;
- Never pour chemicals back into bottles, or exchange stoppers of bottles, because the mixture of chemicals could result in dangerous reactions;
- Never force glass tubing, thermometers or any breakable materials in or out of rubber stoppers and tubing;
- Use glycerine instead of water as a lubricant;
- Laboratory animals, living or dead, should only be handled with instructions from your teacher;
- Solid and other laboratory waste, and broken glass should be placed in designated containers;
- Laboratory work area and equipment should be cleaned and dried before the end of the class;
- Do not **eat** or **drink** anything in the laboratory without instructions from your teacher (even if it is your own drink or food);
- Report any broken glassware to your teacher;

- Use sharp instruments with great care;
- Wash your hands with soap after handling unknown plants or substances;
- When doing field work, make sure that you are safe from animals such as snakes, scorpions and wasps;
- Report any accident or injury to your teacher.

IDENTIFY LABORATORY APPARATUS

- Scientists use special tools or apparatus in their work;
- They also use special materials called **chemicals**;
- Apparatus refer to the set of equipment we use in the lab for experiments

Apparatus and its use	Diagram
Spatulas are used to obtain small samples of solid chemicals. They are also used as a spoon for scooping material out of a container.	
Funnels are used to transfer liquids into other containers with a small neck/ mouth, and for filtering.	

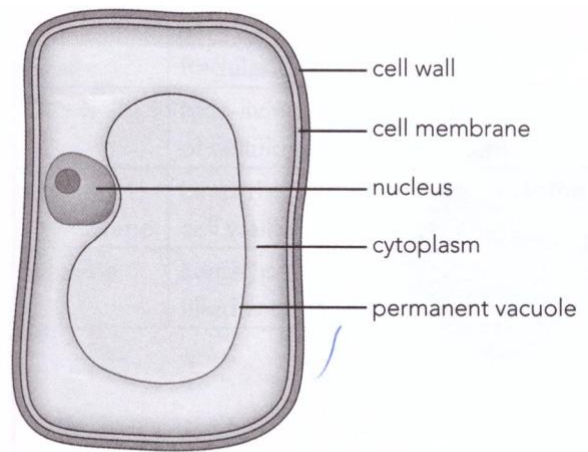
Apparatus and its use	Diagram	Apparatus and its use	Diagram
All these graduated apparatus items can be used to hold specific volumes of liquids. A measuring cylinder is the apparatus used to measure out any volume of a liquid.		Retort stands, bosses and clamps are used to hold the apparatus in position when you do experiments.	
Test tubes hold small amounts of chemicals.		Gauze is placed under a container holding a liquid that is being heated by the Bunsen burner, so that the container does not have direct contact with the flame.	
Test tube racks hold test tubes safely.		Bunsen burners are used to heat substances in the laboratory.	
Crucibles and covers are ceramic containers capable of withstanding extreme temperatures.		Spirit burners are used as alternative means of heating if portable gas and Bunsen burners are not available.	
Evaporation dishes are used for evaporating solutions.			
Tripod stands have three legs and are used to support various flasks during an experiment.			

8.1.2. CELLS AND LEVELS OF ORGANISATION

DESCRIBE THE STRUCTURE OF A PLANT AND AN ANIMAL CELL

The Structure of A Plant Cell

- All plant cells have the following organelles:
- Cell Wall;
- Cell Membrane;
- Cytoplasm;
- Nucleus;
- Permanent Vacuole



Plant cell

Cell Wall: made of a tough substance called cellulose/ is rigid to give the cell its shape and also protects the inner contents of the cell;

Cell Membrane: is a thin living membrane which controls what goes into, or comes out of, the cell/ is a semi-permeable, because it only allows certain substances to pass in and out of the cell;

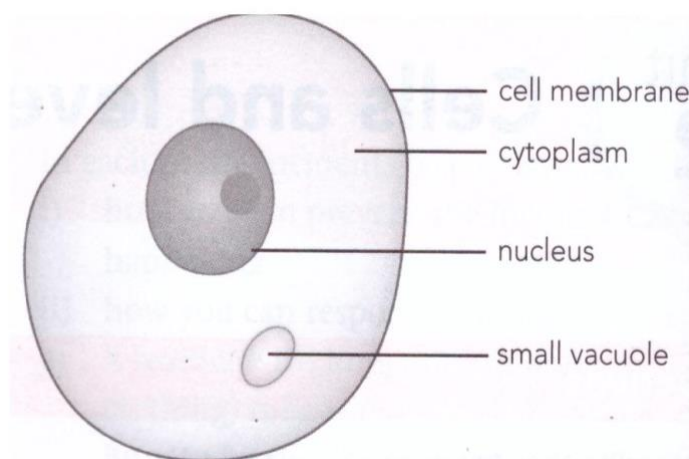
Cytoplasm: is a jelly-like substance in which the organelles float/ most cell activities occur here/ cytoplasm moves around to distribute food and oxygen around the cell;

Nucleus: is a large, round organelles that controls all the activities of the cell, such as growth, making proteins and cell division/ main function is to carry the hereditary information of the cell/ also contain information that controls the chemical reactions in the cell;

Vacuole: are storage area that contain a liquid called sap/ main function is to transport and store water, salts,wastes and food particles

The Structure of An Animal Cell

Animal Cells share similarities with plant cells, but they are also differences between animal and plant cells.



Animal cell

Cell membrane; a thin, living cell membrane, which is flexible, so the cell can become any shape/ controls what enters or leaves the cell/ semi-permiable because it only allows certain substances to pass in and out of the cell;

Cytoplasm: is a jelly-like substance that allows the movement of food and oxygen around the cell/ most cell activities occur here;

Nucleus: controls all the cell activities such as growth, cell division, making proteins, respiration, etc.

DIFFERENCES BETWEEN PLANT AND ANIMAL CELLS

	Plant cell	Animal cell
Shape	Regular	Irregular
Cell wall	Non-living, made of cellulose	No cell wall present
Cell membrane	Lines the inside of cell wall	Surrounds the cell
Vacuole	Large, permanent; filled with cell sap	None or very small

8.1.3 NUTRITION

DEFINE BALANCED DIET

Diet: is the amount and type of food intake that an individual eats and drinks.

Balanced Diet: is the one that provides nutrients in the correct amount and proportions to keep the body healthy.

LIST COMPONENTS OF BALANCED DIET

- A balanced diet includes the following components (nutrients):
 - **Carbohydrates;**
 - **Proteins;**
 - **Lipids (fats and oils)**
 - **Vitamins;**
 - **Mineral Salts;**
 - **Fibre (roughage);**
 - **Water.**

NAME FUNCTION OF NUTRIENTS

- **Carbohydrates:** are sugars (glucose) and starch.
Examples of Foods Rich in Carbohydrates are: sugary and starch foods, such as potatoes, rice, mealie meal, bread, pasta, cereals, and some fruits and vegetables.



Food rich in Carbohydrates

Function: provides the body with energy.

- **Proteins:** are made up of amino acids.
Examples of Foods Rich in Proteins are: meat, fish, eggs, soya, beans, dairy products, mopane worms and nuts.



Foods rich in Proteins

Function: for building body components, repair broken down tissues and growth.

- **Fats and Oils:** contain a higher energy source than Carbohydrates.
Examples of Foods Rich in Fats and Oils are: avocados, nuts, cheese, butter and margarine.



Foods rich in Fats and Oils

Function: reserve source of energy, contain fat-soluble vitamins, keeps the body warm (insulated) and protect internal organs such as kidneys, heart, etc.

- **Vitamins:** are of different types, that is; Vitamin A, Vitamin B, Vitamin C, Vitamin D, Vitamin E and vitamin K.

- o Fat-soluble Vitamins are Vitamin A, D, E and K that are stored in the liver
 - o Water-soluble Vitamins include Vitamin C and Vitamin B.

Examples of Foods Rich in Vitamins A, E and K are: green, leafy vegetables, milk and dairy products, and plant oils / Source of Vitamin D are: fish-liver oils, liver, milk, butter, fish and liver oils

Examples of Foods Rich in Vitamin B are: green, leafy vegetables / Sources of Vitamin C are citrus fruits, green peppers, tomatoes, broccoli, dark leafy vegetables, soya beans and paw paws.

Function: Vitamins control and regulate cellular functions and are important for healthy skin, bones, teeth and gums, and help fight infection.

- **Mineral Salts:** include calcium, phosphorus, magnesium, sodium and potassium.

Examples of Foods Rich in Mineral Salts are: fresh fruits and vegetables, dairy products, meat and eggs

Function: Sodium and Calcium are used to help nerves and muscle to function properly / Calcium also helps blood to clot and strengthens bones and teeth / Iron forms part of the haemoglobin molecule that carries oxygen in the red blood cells.

- **Fibre:** is the indigestible part of plant food found in cereals, such as oats and bran. Does not contain any nutrients.

Examples of Foods Rich in Fibre (Roughage) are: bread, potatoes, fruits, cereals, vegetables and peas.



Foods rich in Fibre (Roughage)

Function: fibre absorbs water, provides bulk to faeces and also absorbs poisonous substances, which are passed out of the body with the faeces.

- **Water:** is very important because it forms a large part of the blood and tissue of the body. It is essential that you drink at least 8 glasses of water every day to replace water that is lost through sweat and urine.

Function: water softens food so that it can move along your alimental canal easily/ transports nutrients and flushes toxins and wastes from the body/ also plays an important role in regulating body temperature that is water is released as sweat to cool body down

8.1.4 RESPIRATORY SYSTEM

STATE THE PERCENTAGE COMPOSITION OF AIR

Composition of Air:

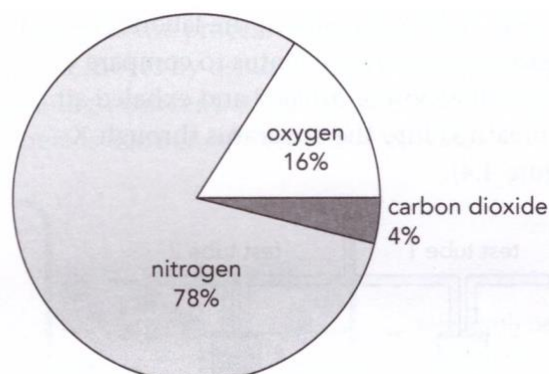
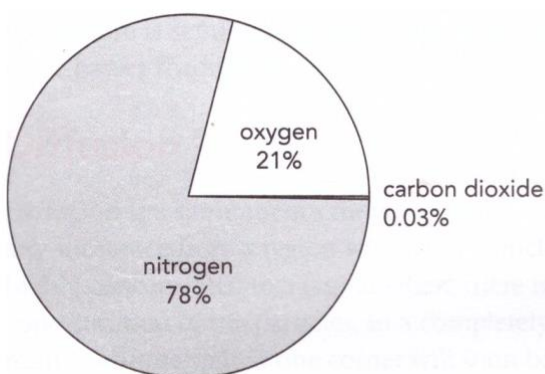
- Oxygen = 21 %
- Carbon dioxide = 0.03 %
- Nitrogen = 78 %
- Water Vapour = Varies (Humidity).

IDENTIFY RESPIRATORY GASSES

- Respiratory gasses are: Oxygen and Carbon dioxide
- During respiration Oxygen is taken in (inhaled when breathing in) and Carbon dioxide is taken out (exhaled when you breathe out).

Gas	Inhaled air (%)	Exhaled air (%)
Oxygen	21	16
Carbon dioxide	0.03	4
Nitrogen	78	78
Water vapour	a small percentage	a larger percentage

Respiration: is the process of burning food in the body cells to produce carbon dioxide, water and release energy.



DESCRIBE THE TEST FOR CARBON DIOXIDE GAS

Carbon dioxide: is a gas produced when fuels burn.

- It is also produced during respiration in plants and animals;
- During respiration, glucose is broken down into carbon dioxide, water and energy.

Experiment 4.2

Aim: To test for the presence of carbon dioxide

Materials: small beaker or test tube, drinking straw, lime water or bicarbonate indicator

Procedure

1. Pour 10 ml lime water into each test tube or beaker.
2. Put the straw into the test tube.
3. Gently blow air from your lungs through the lime water.

Observation

Record the colour changes as exhaled air bubbles through the lime water.

Conclusion

Carbon dioxide turns lime water milky and the red bicarbonate indicator yellow.

Question

1. Using the information given, is carbon dioxide present or absent in exhaled air?

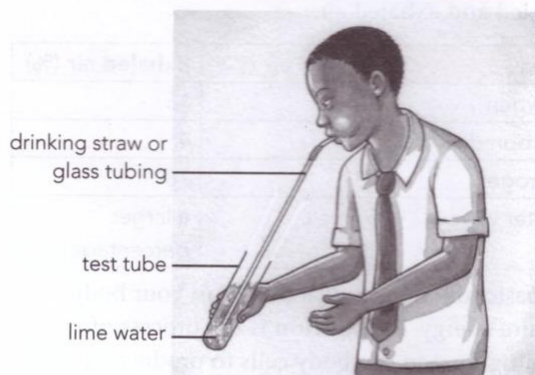


Figure 4.2 Testing for carbon dioxide

TEST FOR OXYGEN GAS

Oxygen: is an important gas in the air;

- We breathe in oxygen to use it for respiration

Experiment 4.1

Aim: To test for oxygen

Materials: manganese dioxide, hydrogen peroxide, wooden splint, a jar of oxygen, matches

Procedure

1. Your teacher will prepare some oxygen for you in the laboratory using manganese dioxide and hydrogen peroxide.
2. Keep the gas jar containing the oxygen tightly closed with a cover.
3. Light the wooden splint and allow it to burn for a short time.
4. Put out the flame.

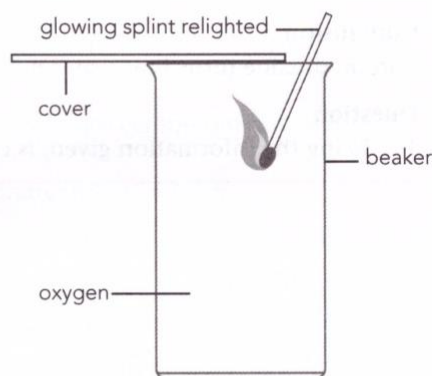


Figure 4.1 Testing for oxygen

5. While the splint is still glowing, put it into the gas jar containing oxygen.
6. Quickly remove the splint and close the jar tight.
7. Put out the flame once more and put the glowing splint back into the gas jar.

Results

Oxygen relights a glowing splint.

Conclusion

This shows that oxygen is present in the jar.

Questions

1. Describe what happens when the glowing splint is placed in the gas jar containing oxygen.
2. Explain why the glowing splint relights in the oxygen in the gas jar.

8.1.5 TRANSPORT SYSTEM

DESCRIBE WATER MOVEMENT IN PLANTS

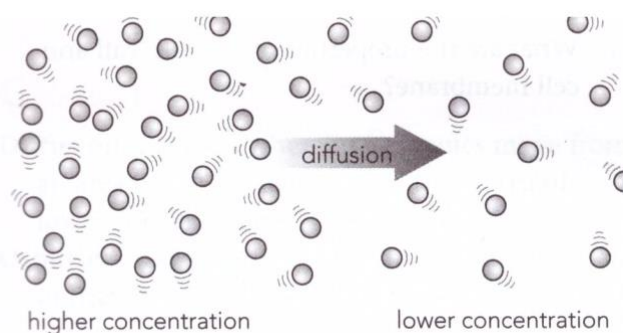
- In plants, molecules of water, nutrients, gasses and waste substances move in and out of cells all the time;
- Substances are moved in and out of the cells by **diffusion** and **osmosis**.

DIFFUSION

Diffusion: is a spontaneous movement of particles of any substance from a region where the particles are highly concentrated to a region where there is low concentration of the particles.

- E.g. perfume can spread from one corner of a closed room with any electric fan to push the smell to another corner due to **diffusion**.

Process of Diffusion

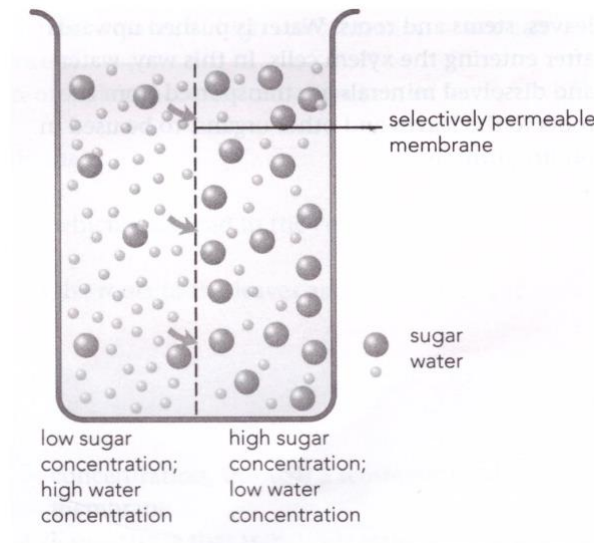


OSMOSIS

Osmosis: is the movement of water molecules from a region of high concentration to a region of lower concentration through a semi-permeable membrane.

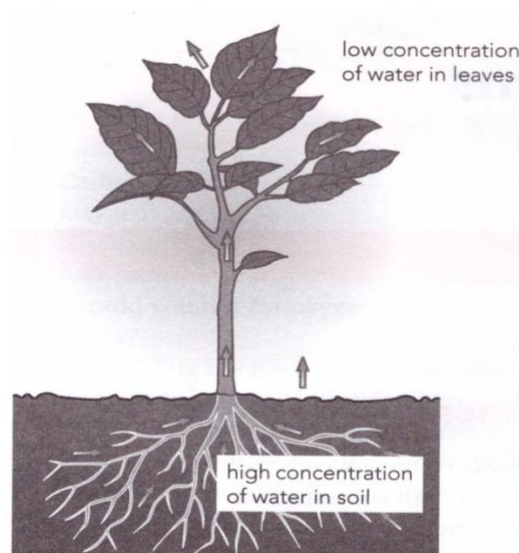
- Water moves through a semi-permeable membrane until the water particles are spread out evenly;
- Osmosis therefore stops when the water particles are in equilibrium.
- E.g. water moves by osmosis from the soil, where there is a high concentration of water into roots, where there is lower concentration of water.

Process of Osmosis

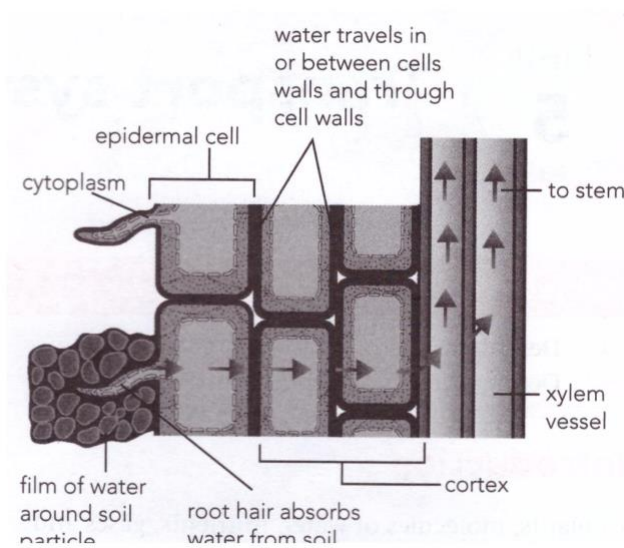


Water Movements in Plants

- Water enters the roots by osmosis;
- The water moves across the root into the **xylem** tissue, located in the centre of the root;
- Xylem carries water throughout plants



Movement of water into the root by osmosis



Roots absorb water by the process of osmosis

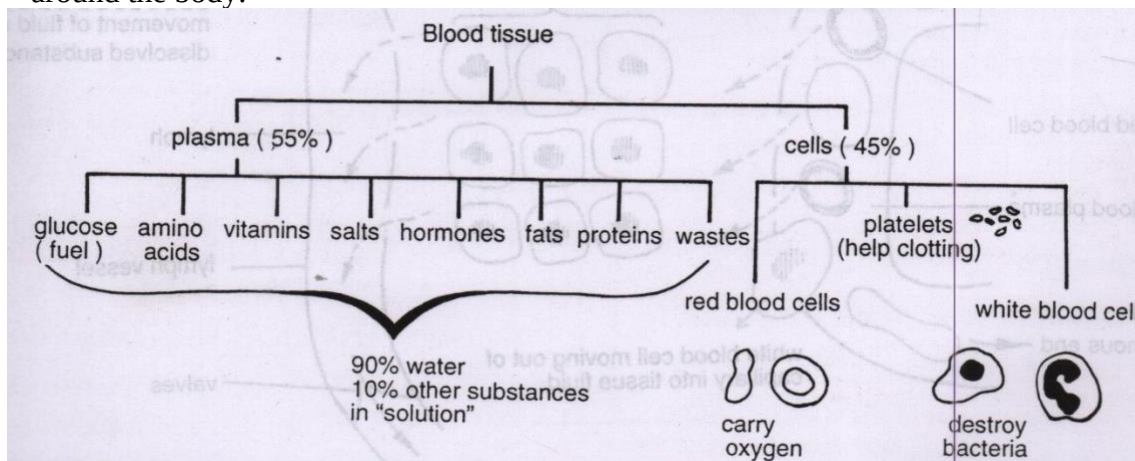
IDENTIFY COMPONENTS OF BLOOD STATING THE FUNCTIONS

- Components of blood are:

- Red blood cells;
- White blood cells;
- Platelets;
- Plasma.

- Functions of the components of blood

- **Red blood cells:** carry oxygen from the lungs to all the cells of the body;
- **White blood cells:** defensive system of the body against diseases, they kill bacteria, virus, fungi, etc. / they form the immunity system of the body.
- **Platelets:** helps in the clotting of blood when injured. Stops bleeding by clotting and avoids too much loss of blood which might lead to death
- **Plasma:** is the liquid part of the blood which carries blood cells, food and waste products around the body.



8.1.6 REPRODUCTION IN PLANTS AND HUMANS

DESCRIBE STRUCTURE OF A SIMPLE FLOWER

- A typical flower has its parts arranged in **four** rings or **whorls**;
- These whorls are attached to a wide base at the end of the flower stalk;
- This wide base is called the **receptacle**;
- The outer whorl is made up of **sepals**;
- The sepals protect the flower bud;
- The **petals** are usually colourful and attract **pollinators**;
- The two inside whorls contain the male and female structures of the flower, which produce the sex cells (gemetes).

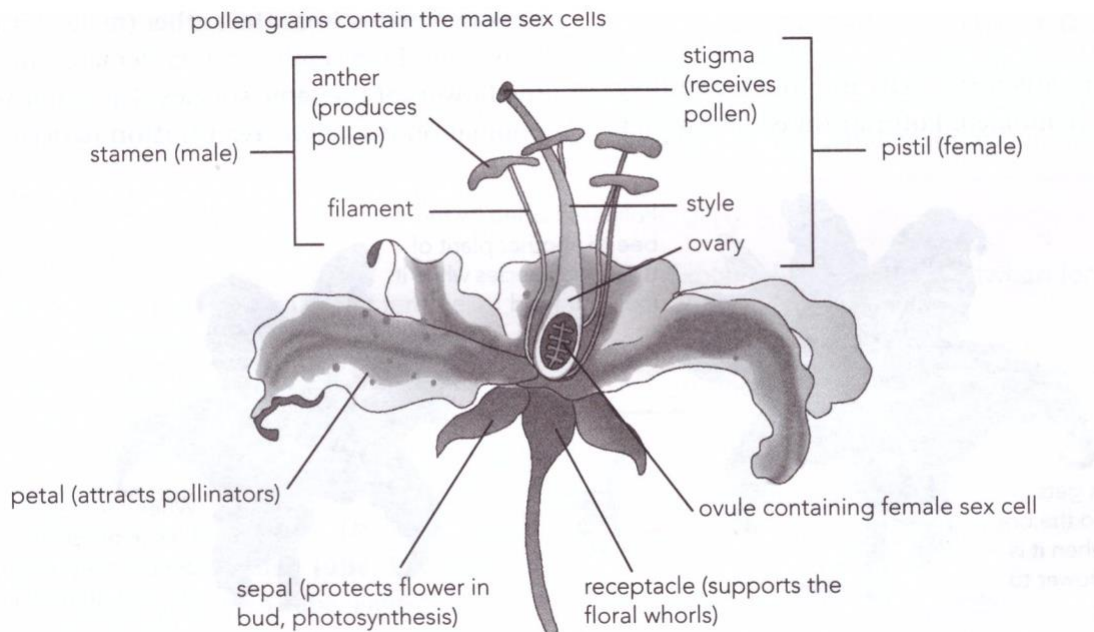
Male Structure

- The male structure is called the **stamen**, which is formed by the **anther** and **filament**;
- The anther is the male structure that produces **pollen grains**, containing male sex cells;
- The anther is held up by the **filament**.

Female Structure

- The female structure is called the **carpel** or **pistil**, and it consists of three parts, the **stigma**, the **style** and the **ovary**;
- The stigma receives pollen, while the ovary produces structure called **ovules**;
- Each ovule holds one female sex cell.

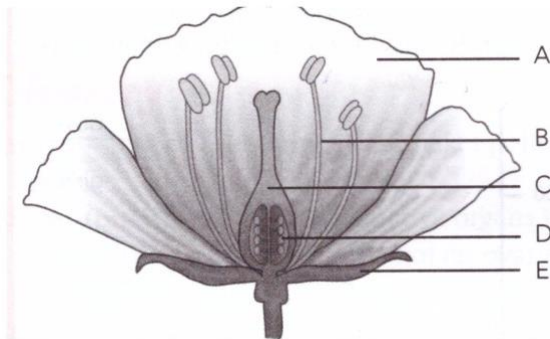
A Simple Flower



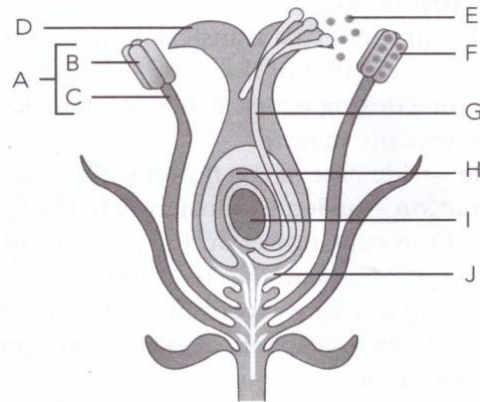
Simple Exercise

Label the following flowers:

a)



b)

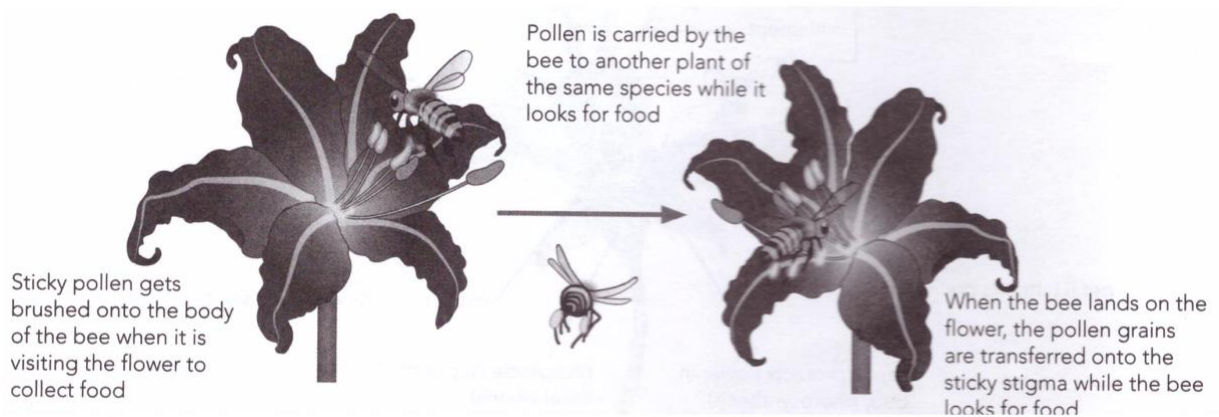


DESCRIBE POLLINATION AND FERTILIZATION

Pollination

- Flowers can only form seeds and fruits when they have been pollinated;
 - o **Pollination:** is the transfer of pollen grains from the anther (male part of the flower) to the receptive stigma (female part of the flower) of the same **species**.
- The purpose of pollination is to allow **fertilization** to occur.
- Pollination can happen in two ways;
- When pollen is being transferred to the stigma of the same flower, it is called **self-pollination**;
- When the pollen is carried to the stigma of another flower, it is called **cross-pollination**
 - o **Pollinator:** such as an insect, bird or animal, carries the pollen from the anther to the stigma of another flower of the same species.

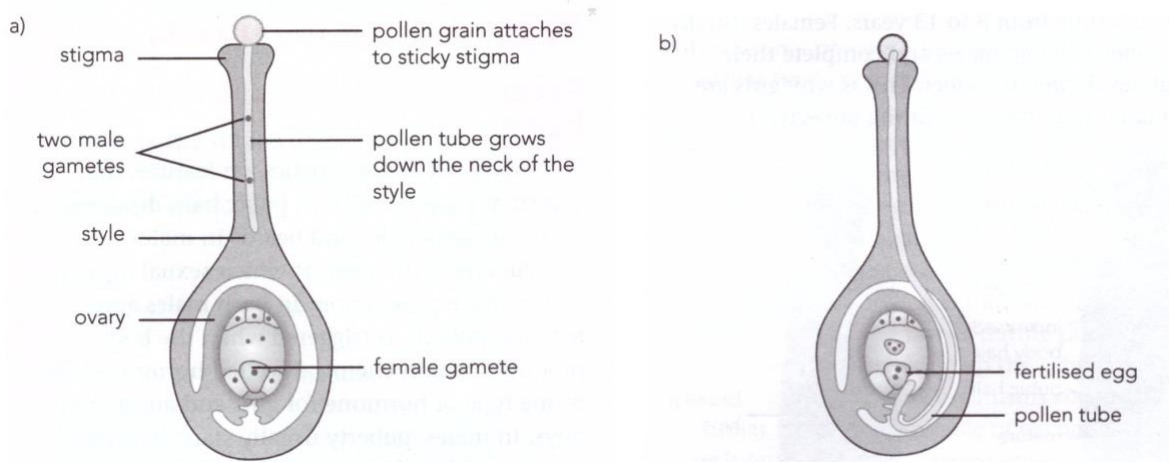
The Process of Pollination



Fertilization

- **Fertilization**: is the fusion/ joining / union of male and female.
- During pollination, many pollen grains land on the stigma of a flower of the same specie;
- A mature pollen grain contains two male gametes;
- The pollen grain bursts open and grows a **pollen tube**;
- The pollen tube grows into the ovule, where one male sex cell fuses with the female egg cell;
- The fertilized egg (ovule) will then develop into a seed;
- The ovary develops into a fruit.

Stages of Fertilization



STATE SIGNS OF PUBERTY

- Puberty is the stage at which sexual organs mature for reproduction;
- In both male and female, puberty is triggered when the body produces certain chemicals called hormones;
- There is one type of hormone for girls and another for boys;
- In males, puberty usually starts between the age of 9 and 14 years, while in female it can begin anytime from 8 to 13 years;

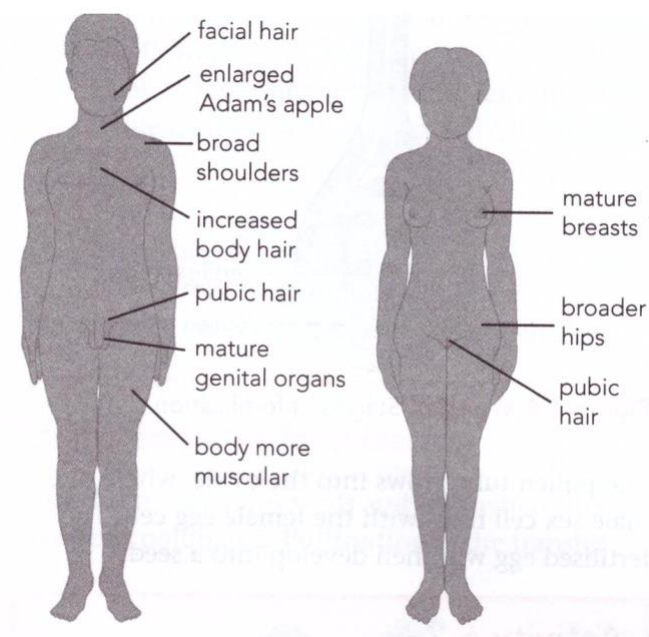
Signs of Puberty in Males:

- Growth of facial, body and pubic hair;
- Increased muscle mass and height;
- Voice changes and becomes deeper;
- Lengthening and widening of the penis

Signs of Puberty in Females:

- Enlarge breast;
- Growth of pubic hair;
- Wider hips;
- Beginning of menstruation;

Changes that Occur at Puberty in Males and females



8.1.7 HEALTH AND DISEASES

DESCRIBE A HEALTHY PERSON

- **Health**: is a state of **mental**, **social** and **physical** well-being;
- **Mental well-being:**
 - To be mental well, you must realize your own abilities and be able to cope with normal stress of life;
 - You must be able to think and work in a productive way when doing school work or other tasks;
 - You can interact with other people and enjoy life;
 - When you are not coping at home or school, it causes mental stress, which means that you are mentally unhealthy.
- **Social well-being:**
 - Social well-being means having friends and seeing them on a regular basis;
 - Personal contact with friends has a positive effect on your mental health and social life;
 - It is important to have friends that you enjoy spending time with and with whom you share a close relationship;
 - Those connections create a positive effect on your health and happiness;
 - To have close personal friends with whom you can laugh, cry and share, increase your happiness and therefore your social well-being.

- **Physical well-being**

- Physical well-being is when all your normal body functions are well coordinated;
- Health may be poor or good;
- Good health means being free from diseases and have a balanced diet;
- Poor health is when you allow certain opportunistic diseases to disturb normal body functions, e.g. lack of Vitamin A will affect your vision/ lack of Vitamin C causes bleeding gums and, in severe cases, scurvy;
- It is essential at your age to have at least eight hours sleep a night;
- Sleep helps rest your body and reduce stress levels.

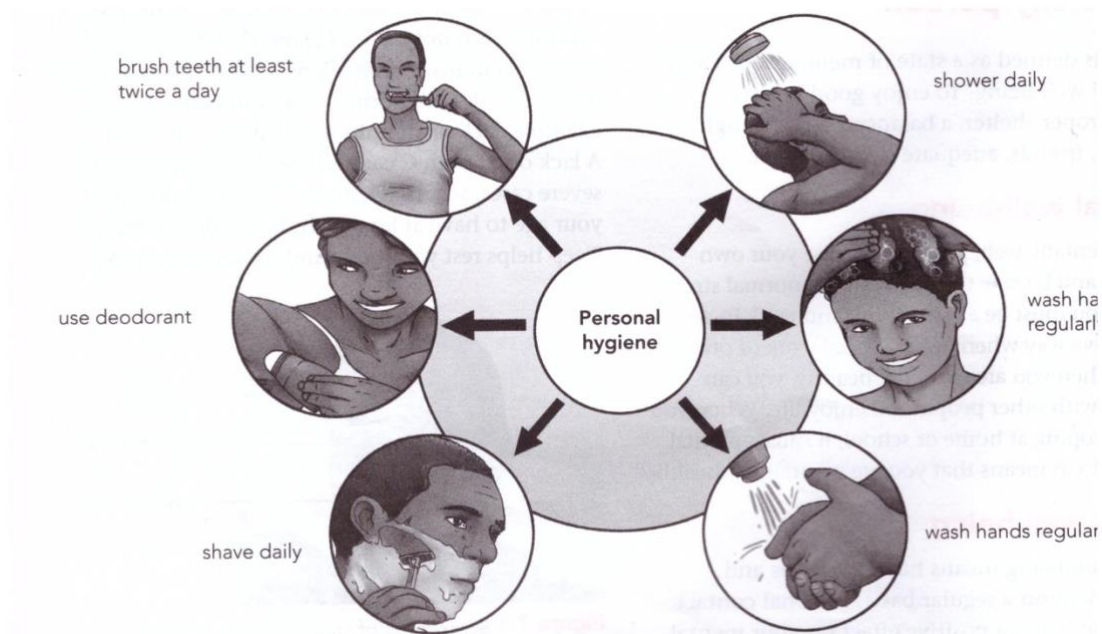
To enjoy good health:

- a person needs proper shelter, a balanced diet, enough exercise, friends, adequate sleep and rest.

STATE IMPORTANCE OF MAINTAINING PERSONAL HYGIENE

- Hygiene refers to conditions and practices that prevent the spread of diseases;
You must make sure that you practice the following personal hygiene habits:
- Bath or wash all over your body with a wet cloth using soap everyday/ soap will remove the dirty and stale sweat/ bacteria will grow in stale sweat, which causes you to smell;
- Wash your hair with soap or shampoo at least once a week/ clean your teeth twice every day to avoid tooth decay/ always wash your hands with soap and water after using the toilet;
- Wash dirty clothes with laundry soap and hang them in the sun to dry/ the sun's rays will kill some **pathogens** (disease-causing germs) and parasites;
- When you iron your clothes, you kill and prevent the growth of lice or fleas, and many other vectors of pathogens;
- Make sure that your home is cleaned at least once a week.

Ways to Practice Personal Hygiene



LIST WAYS OF DISPOSING LITTER, WASTE, STATING THEIR ADVANTAGES AND DISADVANTGES

Ways of Disposing Litter and Waste:

- Burying
- Recycling
- Burning

Burying

- Disposing of solid waste in managed landfill sites and rubbish dumps is the cheapest form of solid waste disposal;
- In landfill and rubbish dumps, waste is first deposited and spread in thick layers;
- It is then compacted by bulldozers;
- The compacted layers are covered with soil;
- The soil prevent bad odours that emanate from decomposing waste from polluting the air;
- Burying waste also prevent insects and rodents from breeding in the waste;
- If not managed well and poorly controlled: harmful substances create a serious health risk to human and animals/ can attract disease-carrying animals, such as rats and insects/ toxic chemical drain into underground water;

Recycling

- Waste that is dumped could usually have been recycled in some way;
- Plastic, glass, metals, paper and cardboard can be recycled and re-moulded into the same or other manufactured goods;
- These materials are collected and transported to recycling plants;
- By recycling, there is less demand for new raw-materials, so we can protect Earth's natural resources;
- Recycling is cheaper than finding suitable land to manage solid waste disposal or controlling illegal rubbish dumps;
- The other advantage of recycling solid waste are that it creates job opportunities for people who collect and recycle waste, and less waste is sent off the landfill sites and incinerators, which releases less air pollution.

Burning

- Some cities produce massive dumps where waste is burnt;
- Incinerators burn household, industrial and organic waste in urban areas;
- Some hospitals dispose of medical waste in their own incinerators;
- Incinerators release waste gases, such as carbon dioxide, water, and oxides of sulphur and nitrogen, as well as ash dust;
- Incinerators are expensive to run and to control emissions into the air;
- People living close to incinerators are vulnerable to lung illness because they breath in toxic gases and particles that are suspended in the air;
- There are also advantages to using incinerators for waste management;
- By burning waste, the amount of land that would have been used for a landfill site is reduced;

- As incinerators are located close to the place where the waste is generated, it saves money on transporting waste;
- When waste is burnt, it prevents the chemicals and toxins leaching into the groundwater, as it does in landfill sites;
- All disease pathogens in the waste are destroyed by high temperatures

DESCRIBE METHODS OF TRANSMISSION OF DISEASES

- Infectious diseases can be transmitted from one infected person to another;
- Staying clean helps to avoid infections;
- An infection occurs when pathogens get into your body and make you sick;
- A pathogen is a disease-causing germ, such as a virus, bacterium or protest;
- Another name for an infectious disease is a communicable disease;
- Communicable diseases are spread in various ways as shown below:

Common Pathogens and infectious Diseases

Type of patho-gen	Examples of infectious diseases
Bacteria	Cholera, tuberculosis (TB), typhoid fever
Protists	Malaria, sleeping sickness
Fungi	Thrush, athlete's foot, ringworm
Viruses	HIV, polio, influenza, chicken pox, Ebola
Parasitic worms	Guinea worm disease, bilharzia

Methods of Transmission of Diseases:

- **Food and Drinking Water**
 - Some micro-organisms, such as *Vibrio cholera* and *Entamoeba histolytica*, get into our drinking water and infect many people;
 - These micro-organisms cause diseases known as **cholera** (*Vibrio cholera*) and **amoebic dysentery** (*Entamoeba histolytica*).
 - These are water-borne diseases;
 - They are common in areas with inadequate sanitation and water purification facilities;
 - An infected person can pass thousands of bacteria in their faeces;
 - This get into the drinking water;
 - You can also get cholera by eating food that has been in contact with contaminated water, flies or dirty hands;
 - A person who has cholera will suffer from diarrhea and vomiting/ this leads to dehydration;
 - If body fluids and electrolytes are not replaced, the person wil fall into a coma and eventually die;
 - To prevent spread of cholera, good food hygiene is essential that is use only proper toilet facilities, wash your hands with soap and water after use, etc.

- **Person to Person Contact**

- Person-to-person contact happens when an infected person touches or exchange bodily fluids with another person;
- Infectious diseases, such as Ebola are caused by deadly viruses

Ebola virus seen under a microscope



- Ebola is transmitted when a person touches broken skin of an infected person or comes into contact with the bodily fluids, such as blood, saliva, sweat, vomit, semen and breast milk of someone infected with the virus;
- Even when someone touches a person who has died from Ebola, they can be infected;
- To prevent Ebola from spreading, make sure you wash your hands with soap and water or an alcohol-based hand sanitizer;
- You should cover your nose and mouth when coughing, and sneeze using a handkerchief, tissue or your sleeve;
- Avoid contact with blood and bodily fluids.

- **Vector-borne Diseases**

- Some diseases are transmitted by **invertebrates**, such as insects (*mosquitoes and fleas*), arachnids (*ticks*) and crustaceans

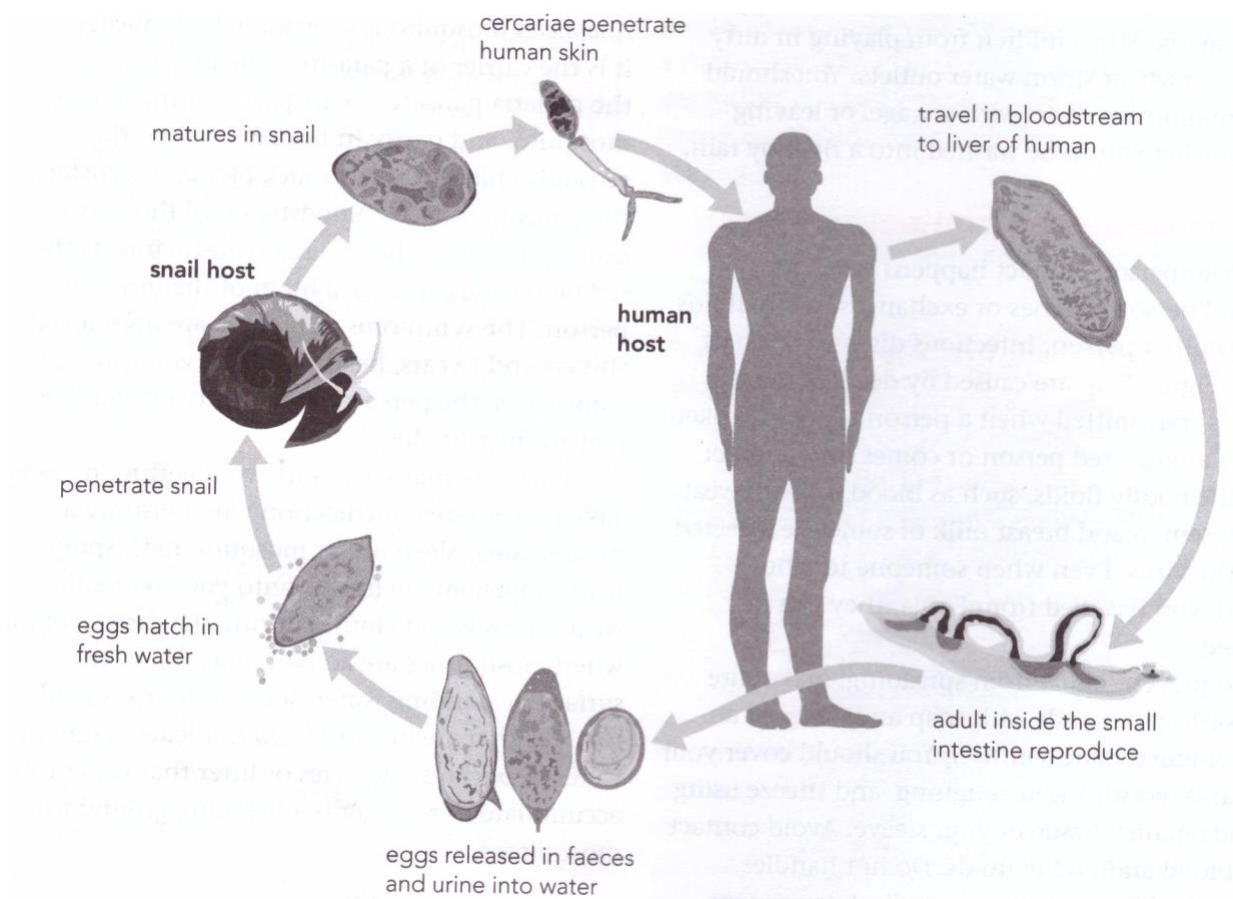
Malaria

- **Malaria:** is caused by a parasite called the *plasmodium* living in the salivary glands of a female *Anopheles* mosquito;
- The *Anopheles* mosquito is a vector, which means it is the carrier of a parasite;
- The life-cycle of the malaria parasite occurs partly in the female mosquito. And partly in humans;
- When the mosquito bites people to suck blood, it transfers the parasite into the bloodstream of the person, causing malaria;
- The malaria parasite infects the red blood cells, liver and brain of the infected person;
- The symptoms of malaria are; **high fevers, shivers and sweats, headaches and vomiting.**
- In some cases, the person can fall into a coma and may eventually die;
- They are ways to avoid getting malaria;
- Take preventative medication when visiting a malaria area;

- Sleep under mosquito nets, spray insect-repellent chemicals onto your skin and wear long-sleeve clothes during the early evening when mosquitoes are active;
- Spray oil on the surface of standing water, such as drains, to kill the lava that are breeding;
- Do not leave stagnant water in buckets, old tyres or litter that water can accumulate in as water is a breeding ground for mosquitoes.

Bilharzias

- **Bilharzia:** is caused by a parasitic flatworm called *schistosoma*.;
- The parasite's life cycle occurs in a freshwater snail and a human;



- The human is the main host for the parasite;
- Bilharzias affects either the urinary system or the intestines and liver;
- The parasite penetrates your skin and enters the blood vessels, where they lay eggs that cause tissue damage;
- Schistosoma lives in streams, dams and rivers;
- People are infected by bilharzias when bathing or drinking water where the snails carrying the worms live;
- It is more common in rural communities where the standards of hygiene are low;
- To prevent infection, do not swim in infected water, such as slow-moving streams and dams;

- Do not urinate or defecate near water used for drinking, washing or bathing and always boil all water before use;
- It is important to create awareness to prevent bilharzias infections.

END-OF-BIOLOGY SECTION REVISION TEST

1. A plant cell is different from an animal cell because it has:
A. a cell membrane
B. a cell wall
C. cytoplasm
D. a nucleus. (2)
2. The part of a cell that can be described as its 'headquarters' is the:
A. vacuole
B. nucleus
C. cytoplasm
D. cell membrane. (2)
3. Exhaled air contains _____ carbon dioxide than inhaled air
A. more
B. less
C. the same amount (2)
4. How would you test for oxygen?
A. By bubbling the gas in lime water
B. Using blue cobalt chloride paper
C. Putting a glowing splint into the gas
D. Putting a burning splint into the gas (2)
5. A learner set up the apparatus in Figure 1 to show that exhaled air contains carbon dioxide. He breathed into the apparatus through X. What observations did he make?

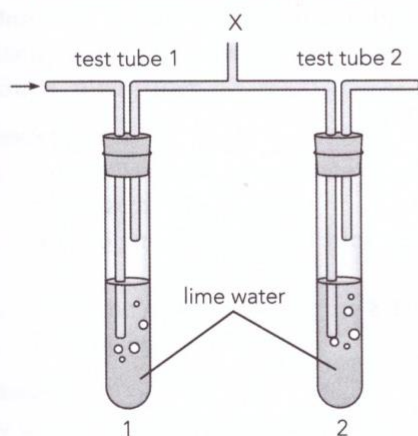


Figure 1 Diagram for question 5

	Test tube 1	Test tube 2
A.	Lime water turns milky	Lime water turns milky
B.	Slight colour change	Lime water turns milky
C.	Lime water turns milky	No colour change
D.	No colour change	No colour change

(2)

6. Why should dam and river water be boiled before drinking?
A. To remove suspended matter
B. To remove dissolved minerals
C. To remove some gases
D. To kill micro-organisms (2)
7. A vector is an organism that:
A. is autotrophic (makes its own food)
B. is most affected by the pathogen
C. carries the disease from one organism to another
D. is heterotrophic (cannot manufacture its own food). (2)
8. Viruses are pathogens because they:
A. cause diseases
B. reproduce within host cells
C. are difficult to control
D. are obligate parasites. (2)
9. An apparatus that holds and keeps test tubes safely is called a:
A. tripod
B. retort stand
C. test tube rack
D. gauze. (2)
10. Scurvy is a disease due to a deficiency of:
A. vitamin D
B. proteins
C. vitamin C
D. fats (2)

11. Figure 2 shows a cell.

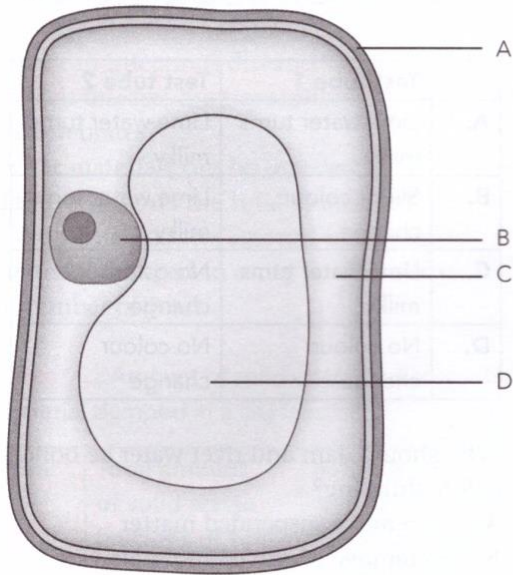


Figure 2 A cell

- Does this cell represent a plant or animal cell? (1)
- Give two reasons for the answer in a). (2)
- Identify organelle B. What is the function of this organelle? (2)
- Tabulate three differences between a plant and animal cell. (4)
- What is the difference between the terms 'permeable' and 'semi-permeable'? Give examples of a structure in a cell that would be permeable or semi-permeable. (4)

12. Study Figure 3 and answer the questions that follow.

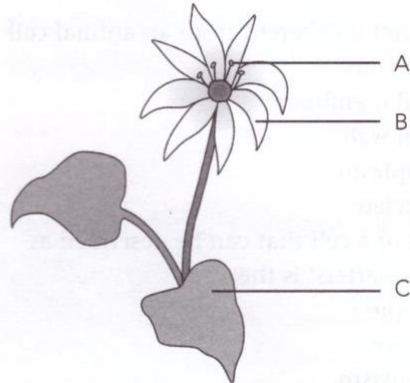


Figure 3 A flower

- Name the parts labelled A, B and C. (3)
 - Give the function of the parts labelled A and B. (2)
 - Define 'pollination'. (1)
 - Explain why plants need to be pollinated. (1)
 - Describe how fertilisation occurs in plants. (4)
13. Tabulate two similarities and two differences of the changes that happen in the body of females and males during puberty. (6)

TOTAL: 50